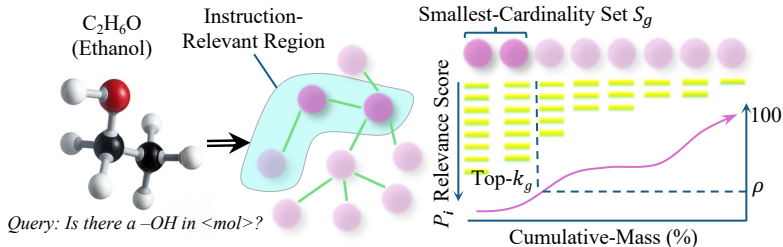


### (a) Smallest-Cardinality Budget Selection



### (b) Bounded Distortion in Patch Growth

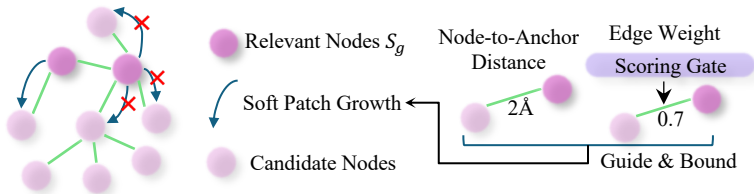


Figure. P1: Illustration of the two components in the adaptive compression analysis. (a) Smallest-cardinality budget selection by sorting node relevance scores and choosing the minimal anchor set  $S_g$  whose cumulative mass reaches the threshold  $\rho$ ; (b) soft patch growth from relevant anchors to nearby candidate nodes, guided by node-to-anchor distance and edge-based scoring.